

From seasons to signals

A macro and systematic guide to FX seasonality

- This note is structured in two parts: Part 1 catalogues all notable G10 FX seasonality patterns and their known or suspected macro drivers. Part 2 presents a systematic approach to constructing weekly seasonality signals out-of-sample, and tests if these signals have held up over time on both a time-series and cross-sectional basis.
- **Main G10 take-aways from the macro lens:** Consistent patterns with credible explanations include **SEK weakness in mid-Mar to Apr** (dividends) and strength in late **Dec** (fiscal year-end), **NOK weakness in Nov** (oil), **CHF strength in Jun/Dec** (bank balance sheet repatriation), and **NZD weakness in Aug** (trade balance low).
- **Main take-aways from the systematic lens:** Trading seasonality systematically has not generated consistent returns over time. Our benchmark signals were generally effective in capturing FX cross-sectional returns pre-2020 (risk-adj returns: +0.37), but have yielded negative returns post-2020 (risk-adj returns: -0.27)...
- ... but we still see merit in tracking pair-specific weekly seasonal effects on a time-series basis, despite the fact that seasonality signals do not work consistently as a cross-sectional factor over time. We highlight the weeks that hold up both pre- and post-2020 as a practical calendar risk map for monitoring specific FX seasonality.

Global FX Strategy

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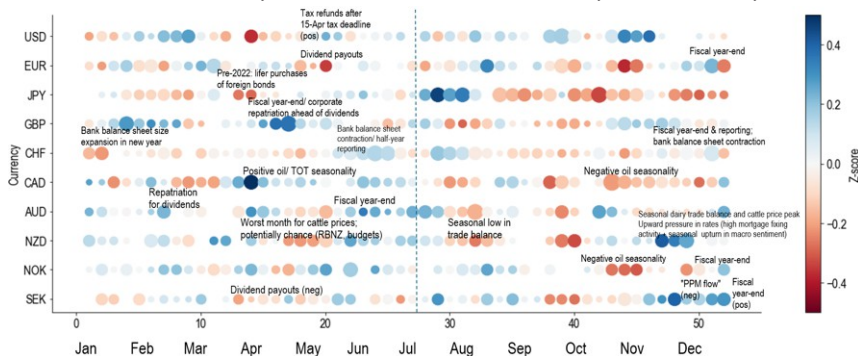
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Figure 1: Some currencies show consistent seasonal patterns with credible explanations, e.g. SEK weakness in Mar/Apr & strength in late Dec, CHF strength in Jun/Dec, and NZD weakness in Aug

Z-score of median NEER performance of past, current & next week; color of dots is z-score (see rhs scale) and size of dots reflects hit rate (% of years where returns are in the direction of the median, 2011-2025). X-axis denotes week and month of the year and vertical line denotes where in the year we are currently.



Source: J.P. Morgan

See page 28 for analyst certification and important disclosures.

Table 1: Table of seasonal patterns that stand out and credible macro explanations

Blue: positive seasonality; red: negative seasonality (for both: darker -> stronger)

Darker green: high hit rate, lighter green/white: low hit rate

Median returns, z-score, hit rates for each period refers to each currency's NEER (trade-weighted index)

Period	Start week	End week	Median return over period (%)	Z-score	Hit rate (%)	Credible explanations
USD Feb	6	9	0.6	0.3	73	-
Early May*	19	19	0.8	0.6	80	Tax refunds after 15-Apr tax deadline; *concentrated vs Asia EM and EUR, SEK, NZD in G10; not broad dollar
Late Sep	39	39	0.4	0.6	87	-
Late Oct-Nov	43	45	0.7	0.5	73	-
EUR Early Jan	1	1	-0.6	-1.0	87	-
Early May*	19	19	-0.8	-0.8	80	Dividend payouts compounded by dollar seasonality
Early March	10	11	0.5	0.5	73	-
Late Dec	52	52	0.3	0.4	73	Fiscal year-end (31-Dec)
JPY Early April	13	13	-0.7	-0.5	73	Previously: lifer purchases of foreign bonds in new fiscal year (which is no longer the case post-2022)
Late Jul	30	30	0.8	0.7	80	-
Early Oct	41	41	-1.2	-1.0	67	-
Nov/Dec	48	49	-1.0	-0.3	73	-
GBP Apr	16	17	0.9	0.8	73	Repatriation of foreign earnings ahead of dividend season
CHF Early Jan	1	2	-0.5	-0.4	80	Reversal in bank balance sheet contraction (after the below)
Jun	22	25	0.8	0.3	87	Seasonal contraction in Swiss bank balance sheet asset size; potential corporate
Dec	52	52	0.3	0.2	80	repatriation ahead of fiscal year-end (31-Dec)
NOK Oct/Nov	42	46	-1.3	-0.6	87	Aligns with negative oil seasonality
SEK Mar/Apr	12	18	-1.0	-0.4	67	Dividend payouts to foreigners
Late Oct	43	45	-0.7	-0.5	73	Chance: dovish Riksbank meetings in last week of October in most years pre-2019
Early Dec	50	50	-0.2	-0.2	67	"PPM flow" by Swedish Pension Agency
Late Dec	51	52	0.9	0.8	80	Fiscal year-end (31-Dec)
AUD Late Feb	7	8	0.4	0.3	73	Repatriation of foreign earnings for dividend season
Mid-Apr	15	15	0.8	0.8	87	-
Aug	31	33	-0.6	-0.3	87	-
Oct	41	43	0.8	0.3	73	-
NZD Apr-May	16	19	-1.5	-0.7	67	-
Mid-May	19	19	-0.6	-0.5	93	Worst month for cattle prices
Aug	31	33	-1.1	-0.6	67	Potentially chance: budget or RBNZ event-driven rather than flows
Late Sep/early Oct	39	40	-0.8	-0.5	80	Seasonal low in trade balance
Late Nov/early Dec	48	49	0.8	0.6	80	Seasonal peak in dairy trade balance and cattle prices
						Potential upward pressure in rates from seasonal upturn in macro sentiment and high point of mortgage re-fixing activity
						Potentially chance: half-year budget update
CAD Late Mar-mid-Apr	13	15	0.9	0.7	80	Positive oil seasonality; potentially gov't fiscal year-end (31-Mar)
Late Oct/early Nov	43	44	-0.6	-0.5	73	Negative oil seasonality

* Median return, z-score and hit rate in this case is for EUR/USD rather than USD NEER (USD early May seasonality)

Source: J.P. Morgan

From seasons to signals

This note examines FX seasonality from two angles – why patterns exist, and whether they have persisted over time and can be traded systematically. It is therefore split into two parts:

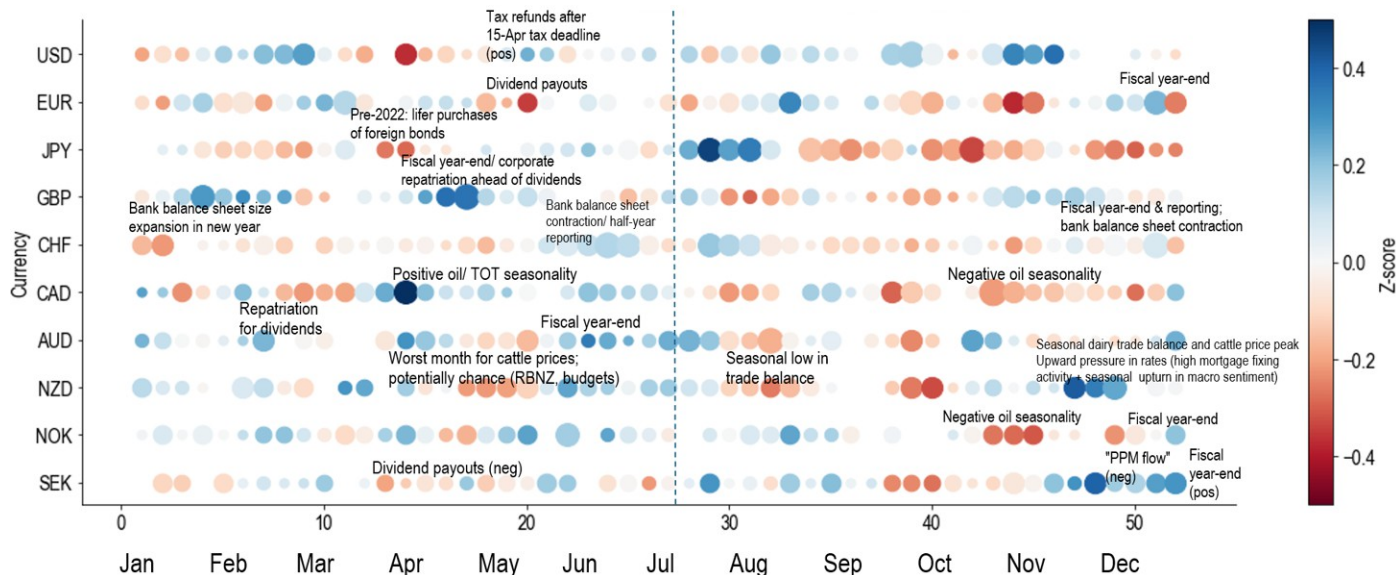
- **Part 1: Macro approach** → we catalogue the main seasonality patterns for G10 FX based on any observed or suspected fundamental drivers;
- **Part 2: Systematic approach** → we construct a standardised seasonal indicator across currencies recursively to test whether these seasonal patterns have been consistent over time, and whether these indicators have worked as effective trading signals. Conclusion: cross-sectional signals worked pre-2020 but not thereafter, but we still see value in tracking individual seasonality patterns, as detailed below.

1. Main take-aways from the macro lens

The **strongest and most consistent patterns observed in G10 FX, which also have credible explanations** are **SEK's Mar & Apr weakness** and **late Dec strength**, **CHF's June and December strength** followed by **early Jan weakness**, **NOK's November weakness**, and **NZD's August and May weakness** and **late November strength**. The section discusses all notable seasonal patterns (judged by higher returns and hit rates) and proposes credible explanations that can be reconciled with the pattern (see details in Table 1).

Figure 2: Some currencies show consistent seasonal patterns with credible explanations, e.g. SEK strength in late Dec, CHF strength in Jun/Dec, and NZD weakness in Aug

Z-score of median NEER performance of past, current & next week; color of dots is z-score (see rhs scale) and size of dots reflects hit rate (% of years where returns are in the direction of the median, since 2011). X-axis denotes week and month of the year and vertical line denotes where in the year we are currently.



Source: J.P. Morgan

2. Main take-aways from the systematic signals

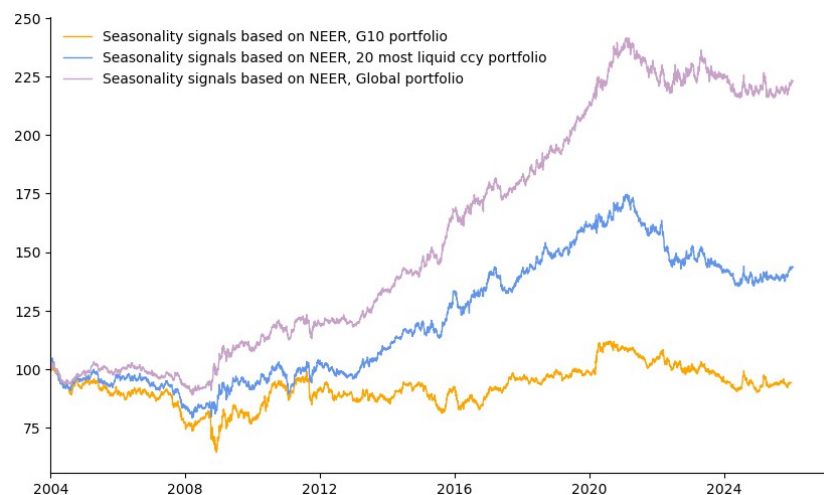
We construct standardised seasonality signals across our usual suite of currencies and conduct both time-series and cross-sectional backtests. This section summarises the main findings; see *Part 2: A systematic approach to seasonality* for full methodology and discussion. Findings are based on our **baseline signal**: the *median of standardised weekly NEER performance over a rolling 10-year window (excluding the current year), smoothed with 50% weight on the current week and 25% each on the prior and next weeks*.

Cross-sectional strategy:

Using seasonality as a trading signal has not delivered consistent performance over time. The signals were quite effective in capturing global FX cross-sectional returns **pre-2020**, delivering risk-adjusted returns of **0.37** (*20 most liquid currencies* portfolio) and **0.74** (*global* portfolio), but have yielded negative performance **post-2020** with **risk-adjusted returns of -0.27** (see Figure 27 for more details).

Figure 3: Seasonality signals delivered consistent performance for the 20 most liquid and Global portfolios (risk-adjusted returns of 0.4-0.7) pre-2020, but have since turned ineffective

Cumulative spot returns of baseline seasonality signals since 2004, top/bottom 2/3/5 on G10/20 most liquid currencies/Global portfolio, ex-trading cost



20 most liquid: EUR, GBP, AUD, NZD, CHF, CAD, NOK, SEK, JPY, USD, BRL, MXN, PLN, ZAR, CNY, INR, KRW, TWD, SGD, THB

Global: EUR, GBP, AUD, NZD, CHF, CAD, NOK, SEK, JPY, USD, BRL, MXN, PLN, ZAR, CNY, INR, KRW, TWD, SGD, THB, CZK, HUF, IDR, MYR, PHP, RUB, TRY, ILS, CLP, COP
Source: J.P. Morgan

Time series strategy:

For each currency pair (either CCY/USD or crosses), we apply the same methodology as in *Part 2: A systematic approach to seasonality* to construct seasonality signals from that pair's weekly performance. The strategy enters a long/short position in the pair when the signal is positive/negative, rebalanced weekly. As in the cross-sectional analysis, results show a **clear regime shift around 2020**, with signals losing relevance post-2020 (for details see *Appendix B: Hit rate of time-series seasonality strategy per currency pair*).

Still, we see merit in tracking **specific seasonal patterns for individual currency pairs that have been consistent both pre- and post-2020**. Figure 4 and Figure 5 below list weeks with significant, persistent seasonalities, screened on four criteria: (1) weekly signal exceeds ± 0.3 sigma (excluding weeks of likely random noise); (2) post-2020 hit rate exceeds 50% in the signal's direction; (3) sufficient trade count (>4 pre-2020 and >2 post-2020); (4) Exception for post-2020: post-2020 hit rate exceeds 75% and trade count > 4 (without pre-2020 constraints). Unlike a static snapshot summarising historical performance, signals here use only information available up to each point in time, providing a more direct out-of-sample test of whether seasonality signals are consistent trading indicators across time.

Comparing against the macro analysis findings (Table 2) we find overlap in some of the most intuitive seasonal windows (e.g., year-end effects and selected flow-related periods), while other weeks are statistically persistent but lack a clear fundamental narrative (highlighted in the "Match with macro findings?" column below). Note that Table 2 in the macro section instead uses trade-weighted indices rather than pairs, includes variable-length seasonal periods, and spans the full 15-year sample rather than the rolling, point-in-time window used here.

Figure 4: Particular seasonal effects are still worth tracking despite seasonality as a cross-sectional factor lost relevance post-2020

Calendar weeks which significant seasonalities are observed for each currency pair both pre-2020 and post-2020 based on the following criteria: 1) weekly signal value should exceed +/- 0.3 sigma (exclude weeks which are likely random noise); 2) hit rate should exceed +50% post-2020 in the direction of the signal; 3) there is a sufficient number of trades occurred over time (>4 trades pre-2020 and >2 trades post-2020); 4) Exception for post-2020: post-2020 hit rate exceeds 75% and trade count > 4 (without pre-2020 constraints)

G10

EUR/USD		pre-2020				post-2020				match with Macro findings?
week	Month	median spot returns	hit rate	number of trades	L/S	median spot returns	hit rate	number of trades	L/S	
19	May	-0.57%	71%	7	S	-1.79%	100%	3	S	Match with USD strength highlighted + EUR dividends
41	Oct	0.12%	50%	4	L	0.57%	100%	2	L	
45	Nov	-1.20%	67%	6	S	-0.16%	50%	6	S	Match with USD strength highlighted Fiscal year end
52	Dec	0.58%	60%	5	L	0.44%	100%	4	L	
GBP/USD		pre-2020				post-2020				match with Macro findings?
week	Month	median spot returns	hit rate	number of trades	L/S	median spot returns	hit rate	number of trades	L/S	
15	Apr	0.33%	69%	13	L	1.18%	100%	2	L	Corporate repatriation around dividend season
16	Apr	1.25%	67%	12	L	0.16%	67%	3	L	
32	Aug	-0.54%	70%	10	S	-0.34%	75%	4	S	
38	Sep	0.25%	64%	14	L	1.18%	50%	2	L	
48	Nov	0.40%	70%	10	L	0.72%	100%	2	L	
AUD/USD		pre-2020				post-2020				match with Macro findings?
week	Month	median spot returns	hit rate	number of trades	L/S	median spot returns	hit rate	number of trades	L/S	
15	Apr	0.93%	86%	7	L	0.47%	50%	2	L	Yes; no macro reasons found
19	May	0.27%	50%	4	S	-0.45%	50%	2	S	
NZD/USD		pre-2020				post-2020				match with Macro findings?
week	Month	median spot returns	hit rate	number of trades	L/S	median spot returns	hit rate	number of trades	L/S	
19	May	-0.89%	100%	11	S	-1.04%	100%	4	S	Yes; unclear macro drivers
36	Sep	-1.34%	60%	5	S	-0.68%	100%	3	S	
48	Nov	-0.03%	50%	8	L	2.05%	100%	2	L	Seasonal peak in dairy trade balance
CHF/USD		pre-2020				post-2020				match with Macro findings?
week	Month	median spot returns	hit rate	number of trades	L/S	median spot returns	hit rate	number of trades	L/S	
39	Sep	-0.75%	43%	7	L	-0.49%	75%	4	S	Seasonal contraction in Swiss bank balance sheet; potential corporate repatriation
44	Nov	-0.39%	45%	11	L	-0.32%	75%	4	S	
45	Nov	-1.06%	67%	6	S	-0.31%	67%	6	S	
48	Nov	1.03%	75%	8	L	0.68%	100%	2	L	
50	Dec	-0.41%	50%	8	L	0.72%	50%	2	L	
51	Dec	0.34%	75%	4	L	0.11%	50%	6	L	
CAD/USD		pre-2020				post-2020				match with Macro findings?
week	Month	median spot returns	hit rate	number of trades	L/S	median spot returns	hit rate	number of trades	L/S	
13	Mar	0.27%	60%	5	L	1.00%	100%	2	L	Trade balance/ oil prices negative oil seasonality
45	Nov	-0.54%	78%	9	S	-0.12%	50%	6	S	
NOK/USD		pre-2020				post-2020				match with Macro findings?
week	Month	median spot returns	hit rate	number of trades	L/S	median spot returns	hit rate	number of trades	L/S	
45	Nov	-0.66%	80%	5	S	-0.03%	50%	6	S	negative oil seasonality
SEK/USD		pre-2020				post-2020				match with Macro findings?
week	Month	median spot returns	hit rate	number of trades	L/S	median spot returns	hit rate	number of trades	L/S	
10	Mar	0.51%	80%	5	L	1.30%	100%	3	L	around dividend payout period
19	May	-0.69%	71%	7	S	-0.43%	100%	2	S	
29	Jul	0.08%	75%	4	L	0.63%	50%	4	L	
45	Nov	-1.06%	50%	6	S	-0.15%	60%	5	S	Yes; unclear macro drivers Fiscal year end
52	Dec	0.23%	62%	8	L	0.61%	100%	2	L	
JPY/USD		pre-2020				post-2020				match with Macro findings?
week	Month	median spot returns	hit rate	number of trades	L/S	median spot returns	hit rate	number of trades	L/S	
28	Jul	0.03%	60%	5	L	0.38%	100%	2	L	Yes; no macro reasons found
37	Sep	-0.98%	80%	10	S	-0.12%	60%	5	S	
49	Dec	-0.29%	57%	7	S	-0.18%	75%	4	S	

G10 crosses

EURGBP		pre-2020					post-2020			
week	Month	median P&L	hit rate	number of trades	L/S	median P&L	hit rate	number of trades	L/S	
4	Jan	-0.45%	40%	10	S	-0.36%	75%	4	S	
9	Mar	0.18%	60%	10	L	0.96%	50%	2	L	
26	Jun	0.49%	50%	10	L	0.38%	100%	4	L	
50	Dec	0.31%	60%	5	L	0.91%	75%	4	L	
EURNOK		pre-2020					post-2020			
week	Month	median P&L	hit rate	number of trades	L/S	median P&L	hit rate	number of trades	L/S	
7	Feb	-0.58%	78%	9	S	-0.25%	67%	3	S	
14	Apr	-0.26%	82%	11	S	-0.34%	50%	6	S	
19	May	-0.89%	83%	6	S	-0.04%	60%	5	S	
44	Oct	0.19%	55%	11	S	0.40%	75%	4	L	
EURSEK		pre-2020					post-2020			
week	Month	median P&L	hit rate	number of trades	L/S	median P&L	hit rate	number of trades	L/S	
10	Mar	0.02%	50%	2	L	-0.74%	75%	4	S	
21	May	-1.06%	100%	4	S	-0.36%	50%	6	S	
AUDNZD		pre-2020					post-2020			
week	Month	median P&L	hit rate	number of trades	L/S	median P&L	hit rate	number of trades	L/S	
28	Jul	2.71%	0%	1	S	-0.45%	100%	4	S	
42	Oct	0.22%	38%	8	S	-0.20%	75%	4	S	
46	Nov	-0.61%	57%	7	S	-0.38%	83%	6	S	
47	Nov	-0.26%	54%	13	S	-0.22%	67%	6	S	
48	Nov	-0.34%	67%	15	S	-0.68%	100%	6	S	

Source: J.P. Morgan

Figure 5: Specific seasonal effects for EM FX (CCY/USD) which appear to be consistent pre-2020 and post-2020 or exhibit significant pattern post-2020

Calendar weeks in which significant seasonalities are observed for each currency pair, both pre-2020 and post-2020, based on the following criteria: 1) weekly signal value should exceed +/- 0.3 sigma (exclude weeks which are likely random noise); 2) hit rate should exceed +50% post-2020 in the direction of the signal; 3) there is a sufficient number of trades occurring over time (>4 trades pre-2020 and >2 trades post-2020); 4) Exception for post-2020: post-2020 hit rate exceeds 75% and trade count > 4 (without pre-2020 constraints)

LatAm

CLP/USD		pre-2020				post-2020			
week	Month	median spot returns	hit rate	number of trades	L/S	median spot returns	hit rate	number of trades	L/S
7	Feb	0.45%	62%	8	L	0.67%	100%	2	L
13	Mar	0.45%	100%	8	L	1.03%	50%	6	L
28	Jul	0.66%	75%	8	L	0.54%	67%	3	L
52	Dec	-	-	0	-	1.90%	80%	5	L

COP/USD		pre-2020				post-2020			
week	Month	median spot returns	hit rate	number of trades	L/S	median spot returns	hit rate	number of trades	L/S
13	Mar	1.25%	100%	2	L	0.87%	80%	5	L
15	Apr	0.66%	60%	10	L	1.21%	67%	6	L
20	May	-1.28%	75%	8	S	-0.51%	67%	3	S
22	Jun	0.47%	62%	8	L	1.15%	100%	2	L
33	Aug	-0.70%	67%	6	S	-2.60%	100%	4	S
46	Nov	-0.89%	75%	4	S	-1.34%	67%	3	S

BRL/USD		pre-2020				post-2020			
week	Month	median spot returns	hit rate	number of trades	L/S	median spot returns	hit rate	number of trades	L/S
19	May	-0.47%	64%	14	S	-0.74%	67%	3	S
41	Oct	0.61%	50%	8	L	0.93%	67%	3	L

MXN/USD		pre-2020				post-2020			
week	Month	median spot returns	hit rate	number of trades	L/S	median spot returns	hit rate	number of trades	L/S
13	Mar	-0.37%	40%	5	L	1.08%	83%	6	L
41	Oct	1.00%	75%	8	L	0.18%	50%	6	L
44	Nov	-0.14%	50%	4	S	-0.14%	50%	2	S

EMEA

PLN/USD		pre-2020				post-2020			
week	Month	median spot returns	hit rate	number of trades	L/S	median spot returns	hit rate	number of trades	L/S
19	May	-0.06%	60%	5	S	-0.83%	50%	2	S
30	Jul	0.34%	60%	5	L	0.71%	75%	4	L
41	Oct	2.66%	100%	2	L	1.02%	100%	5	L
51	Dec	-0.17%	44%	9	L	0.42%	80%	5	L
52	Dec	0.73%	67%	6	L	0.28%	80%	5	L

CZK/USD		pre-2020				post-2020			
week	Month	median spot returns	hit rate	number of trades	L/S	median spot returns	hit rate	number of trades	L/S
39	Sep	-0.47%	45%	11	L	-0.56%	80%	5	S
45	Nov	-1.39%	67%	6	S	-0.13%	50%	6	S

HUF/USD		pre-2020				post-2020			
week	Month	median spot returns	hit rate	number of trades	L/S	median spot returns	hit rate	number of trades	L/S
41	Oct	0.63%	75%	4	L	1.13%	100%	3	L

ILS/USD		pre-2020				post-2020			
week	Month	median spot returns	hit rate	number of trades	L/S	median spot returns	hit rate	number of trades	L/S
39	Sep	-0.89%	14%	7	L	-0.31%	75%	4	S
48	Nov	1.05%	82%	11	L	0.70%	100%	3	L

ZAR/USD		pre-2020				post-2020			
week	Month	median spot returns	hit rate	number of trades	L/S	median spot returns	hit rate	number of trades	L/S
1	Jan	0.00%	67%	6	S	-1.28%	75%	4	S
7	Feb	-1.17%	50%	4	L	0.32%	50%	2	L
29	Jul	0.65%	78%	9	L	0.63%	67%	6	L
41	Oct	1.67%	80%	5	L	0.23%	67%	6	L

Asia

PHP/USD		pre-2020				post-2020			
week	Month	median spot returns	hit rate	number of trades	L/S	median spot returns	hit rate	number of trades	L/S
3	Jan	-	-	0	-	-0.10%	75%	4	S
20	May	-0.25%	88%	8	S	-0.38%	67%	3	S
24	Jun	-0.68%	86%	7	S	-0.92%	83%	6	S
37	Sep	-0.01%	100%	1	S	-0.26%	83%	6	S

INR/USD		pre-2020				post-2020			
week	Month	median spot returns	hit rate	number of trades	L/S	median spot returns	hit rate	number of trades	L/S
19	May	-0.53%	83%	6	S	-0.63%	83%	6	S
24	Jun	-0.34%	100%	5	S	-0.35%	80%	5	S
32	Aug	-0.24%	80%	5	S	-0.22%	100%	6	S
33	Aug	-0.58%	67%	6	S	-0.18%	67%	3	S
46	Nov	-0.59%	71%	7	S	-0.03%	67%	6	S

KRW/USD		pre-2020				post-2020			
week	Month	median spot returns	hit rate	number of trades	L/S	median spot returns	hit rate	number of trades	L/S
13	Mar	1.16%	83%	6	L	1.51%	100%	2	L
19	May	0.47%	40%	5	S	-0.90%	100%	5	S
43	Oct	0.39%	71%	7	L	0.63%	60%	5	L

TWD/USD		pre-2020				post-2020			
week	Month	median spot returns	hit rate	number of trades	L/S	median spot returns	hit rate	number of trades	L/S
17	Apr	-0.02%	50%	6	L	0.05%	67%	3	L
24	Jun	-0.57%	100%	4	S	-0.09%	50%	6	S

SGD/USD		pre-2020				post-2020			
week	Month	median spot returns	hit rate	number of trades	L/S	median spot returns	hit rate	number of trades	L/S
13	Mar	0.22%	75%	4	L	0.08%	100%	2	L
15	Apr	0.14%	57%	7	L	0.41%	80%	5	L
19	May	-0.29%	67%	6	S	-0.54%	100%	6	S
39	Sep	-0.36%	100%	2	S	-0.29%	83%	6	S

THB/USD		pre-2020				post-2020			
week	Month	median spot returns	hit rate	number of trades	L/S	median spot returns	hit rate	number of trades	L/S
3	Jan	0.58%	62%	8	L	0.35%	67%	6	L
19	May	-0.59%	86%	7	S	-0.21%	67%	6	S
39	Sep	-	-	0	-	-0.69%	80%	5	S

Source: J.P. Morgan

Part 1: The macro lens – G10 FX seasonality catalogue

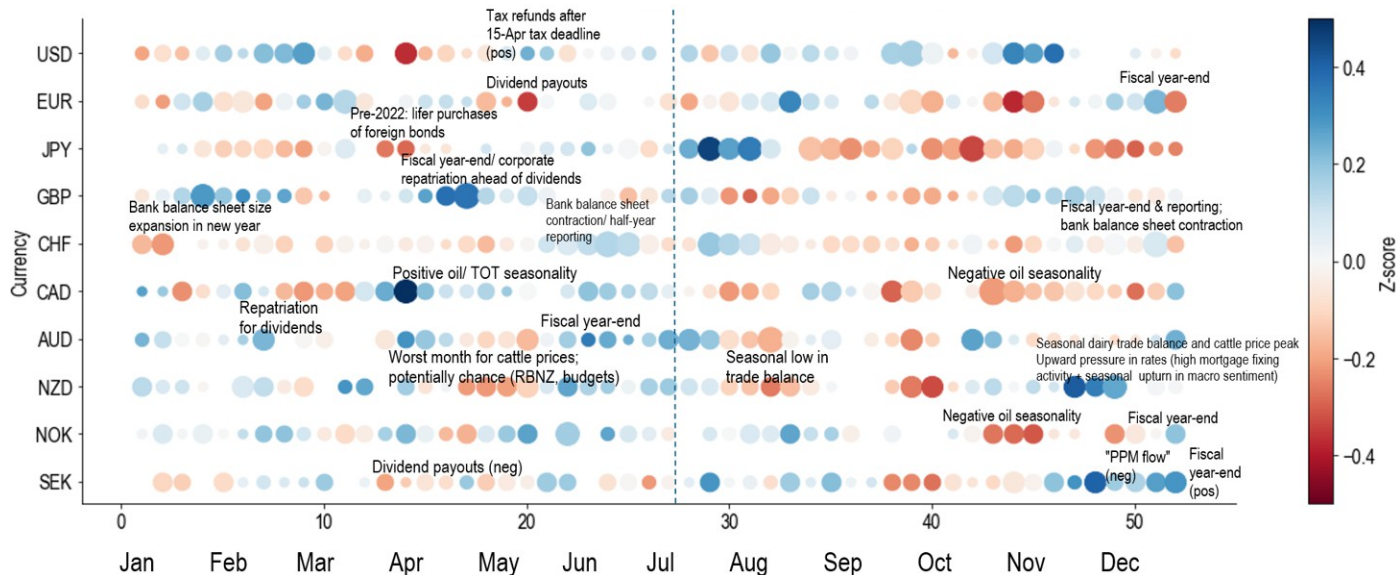
This section discusses notable G10 FX seasonality patterns and their known or suspected drivers. Generally, a potential seasonal pattern can stem from one of the following:

- **Equity dividends** (e.g. SEK in spring; see [Asia FX dividend seasonality](#), Michael Loh);
 - The sign depends on the earnings split and ownership structure: repatriation by domestic firms with large overseas exposure ahead of payouts is local-currency-positive, while foreign investors repatriating dividend payouts is local-currency-negative.
- **Commodity price and trade balance seasonality** (e.g. NOK with oil, NZD with dairy; see Appendix D Figure 30 for commodity terms-of-trade seasonality by currency);
- Seasonal **portfolio flows** or **corporate repatriation** (e.g. around fiscal year-end or reporting season);
- **Risk sentiment** seasonality ('Sell in May and go away');
- ... or finally, **pure chance rather than a fundamental/ flow driver**, e.g. a central bank that happened to surprise dovishly/hawkishly in the same months across several years, or other one-off macro drivers that recurred by coincidence

Seasonal patterns driven by flows are more apparent in lower-liquidity G10 FX and less so in USD and EUR; the **strongest and most consistent patterns observed** are **SEK's mid-March to April weakness**, **CHF's June and December strength** followed by early Jan weakness, **NOK's November weakness**, **NZD's August and May weakness** and late **November strength**, and **AUD's mid-April strength**.

Figure 6: Some currencies show consistent seasonal patterns with credible explanations, e.g. SEK strength in late Dec, CHF strength in Jun/Dec, and NZD weakness in Aug

Z-score of median weekly NEER performance smoothed across past, current & next week; color of dots is z-score of median return (see rhs scale) and size of dots reflects hit rate (% of years where returns are in the direction of the median, since 2011). X-axis denotes week and month of the year and vertical denotes where in the year we are currently.



Source: J.P. Morgan

In more detail (shown also in Figure 6 and Table 2):

- **SEK:**
 - **Weakness** around **March/April** (median of -1% in SEK TWI over mid-March through April; 67% hit rate) driven by **dividend outflows** (Figure 7)
 - **Early December** first sees a known negative flow from annual premium-pension investments, but this is less pronounced/consistent in price action than the subsequent positive seasonality (see [European FX year-end seasonality: Krona takes the crown](#), Popescu).
 - **Second-half-of-December strength** (+0.9% in SEK TWI; 80% hit rate) is likely related to the **fiscal year-end**.
- **CHF:**
 - **June** (median of +0.8% in CHF TWI over June; 87% hit rate) and **December** (+0.3%; 80% hit rate) **strength** and **subsequent January weakness** (-0.5%; 80% hit rate) can be reconciled with **bank balance sheet** (and possibly corporate) **repatriation** around fiscal year-end and reporting (see Figure 8 and discussion in [European FX year-end seasonality: Krona takes the crown](#), Popescu).
- **NZD:**
 - **August weakness** (-1.1% over 3w; 67% hit rate) aligns with **seasonal low in the trade balance** (Figure 9).
 - **Late Nov/early Dec** (+0.8% over 2w; 80% h.r.) **strength** aligns with **seasonal peak in dairy trade balance** and **cattle prices**, plus rates drivers (see Table).
 - **Negative May** seasonality (-1.5% over 4w with 67% h.r. with one single week at -0.6% with a 93% hit rate) and **late September** (-0.8% over 2w; 80% h.r.) also stand out, though with unclear drivers (May may have been event-driven, e.g. RBNZ).
- **NOK:**
 - **November weakness** (-1.3% over 5w; 87% hit rate) aligns with **negative oil seasonality** (Figure 10).
- **JPY:**
 - **Early April weakness** (-0.7% over 1w with 73% hit rate) was **previously** explained by **lifer purchases of foreign bonds** in new fiscal year, but this is no longer the case since 2022 (Figure 11).
 - **Late-July strength** (0.8%; 80% h.r.) as well as **weakness** in **early October** (-1.2%; 67% hit rate) and **late Nov/early Dec** (-1%; 73% hit rate) lack viable explanations.
- **AUD:**
 - **Positive returns in Mid-April** (+0.8% over 1wk with 87% h.r.) and in **October** (0.8% over 3w; 73% h.r.) are observed without apparent cause, but these are concentrated pre-2021.
 - **Late Feb modestly positive seasonality** is associated with **dividend season** (where local corporates pay dividends out of foreign earnings; +0.4% over 2w; 73% h.r.); this should also be the case in Aug/Sep but does not consistently appear in price action.
- **GBP:**
 - **April strength** (+0.9% over 2w; 73%) can be reconciled with corporate repatriation ahead of dividend season.
- **EUR:**
 - **Weakness** in the **first week** of the year (-0.6%; 87% h.r.) which reverses (lower-intensity: +0.3%; 73% h.r.) **strength** in the last week of the year, potentially reflecting **repatriation** and reversal around **fiscal year-end**.

- **Early May weakness** can be reconciled with **dividend season**, accentuated by broader dollar (EUR/USD falls by median of 0.8% in week 19; 80% h.r.; see USD below).
- **CAD:**
 - Notable seasonal patterns align with **oil/terms-of-trade** seasonality: **positive** in **late Mar/mid-Apr** (+0.9% over 3w; 80% h.r.) and **negative** in **late Oct/early Nov** (less pronounced; -0.6% over 2w; 73% h.r.)
- **USD:**
 - Week 19 (**early May**) is a highly-positive week for the dollar **vs low-yielders** (EUR, SEK, NZD in DM and Asia EM) though it does not show up as consistently in the trade-weighted index (but EUR/USD falls by median of 0.8%; 80% h.r.). This can be attributed in part to **tax refunds after the April 15th tax deadline**, and is worsened by local dividend seasons for some (see detail on p.1 and p.28 [here](#)).
 - Otherwise, **late Sep** (+0.4% over 1w; 87% hit rate) and **late Oct-Nov** (+0.7% over 3w; 73% hit rate) stand out for the broad NEER, though without an apparent driver in either.

Table 2: Table of seasonal patterns that stand out and credible explanations

Blue: positive seasonality; red: negative seasonality (for both: darker -> stronger)

Darker green: high hit rate, lighter green/white: low hit rate

Median returns, z-score, hit rates for each period refers to each currency's NEER (trade-weighted index)

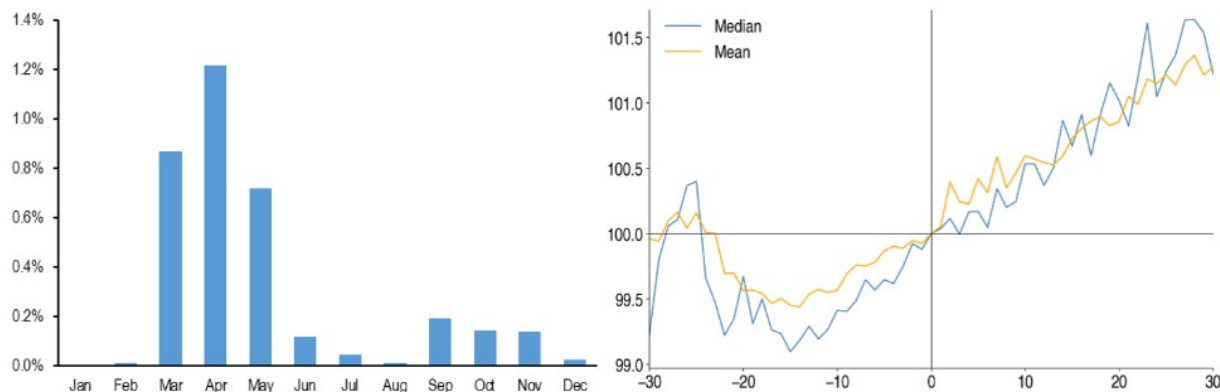
Period	Start week	End week	Median return over period (%)	Z-score	Hit rate (%)	Credible explanations
USD Feb	6	9	0.6	0.3	73	-
Early May*	19	19	0.8	0.6	80	Tax refunds after 15-Apr tax deadline; *concentrated vs Asia EM and EUR, SEK, NZD in G10; not broad dollar
Late Sep	39	39	0.4	0.6	87	-
Late Oct-Nov	43	45	0.7	0.5	73	-
EUR Early Jan	1	1	-0.6	-1.0	87	-
Early May*	19	19	-0.8	-0.8	80	Dividend payouts compounded by dollar seasonality
Early March	10	11	0.5	0.5	73	-
Late Dec	52	52	0.3	0.4	73	Fiscal year-end (31-Dec)
JPY Early April	13	13	-0.7	-0.5	73	Previously: lifer purchases of foreign bonds in new fiscal year (which is no longer the case post-2022)
Late Jul	30	30	0.8	0.7	80	-
Early Oct	41	41	-1.2	-1.0	67	-
Nov/Dec	48	49	-1.0	-0.3	73	-
GBP Apr	16	17	0.9	0.8	73	Repatriation of foreign earnings ahead of dividend season
CHF Early Jan	1	2	-0.5	-0.4	80	Reversal in bank balance sheet contraction (after the below)
Jun	22	25	0.8	0.3	87	Seasonal contraction in Swiss bank balance sheet asset size; potential corporate
Dec	52	52	0.3	0.2	80	repatriation ahead of fiscal year-end (31-Dec)
NOK Oct/Nov	42	46	-1.3	-0.6	87	Aligns with negative oil seasonality
SEK Mar/Apr	12	18	-1.0	-0.4	67	Dividend payouts to foreigners
Late Oct	43	45	-0.7	-0.5	73	Chance: dovish Riksbank meetings in last week of October in most years pre-2019
Early Dec	50	50	-0.2	-0.2	67	"PPM flow" by Swedish Pension Agency
Late Dec	51	52	0.9	0.8	80	Fiscal year-end (31-Dec)
AUD Late Feb	7	8	0.4	0.3	73	Repatriation of foreign earnings for dividend season
Mid-Apr	15	15	0.8	0.8	87	-
Aug	31	33	-0.6	-0.3	87	-
Oct	41	43	0.8	0.3	73	-
NZD Apr-May	16	19	-1.5	-0.7	67	-
Mid-May	19	19	-0.6	-0.5	93	Worst month for cattle prices Potentially chance: budget or RBNZ event-driven rather than flows
Aug	31	33	-1.1	-0.6	67	Seasonal low in trade balance
Late Sep/early Oct	39	40	-0.8	-0.5	80	-
Late Nov/early Dec	48	49	0.8	0.6	80	Seasonal peak in dairy trade balance and cattle prices Potential upward pressure in rates from seasonal upturn in macro sentiment and high point of mortgage re-fixing activity Potentially chance: half-year budget update
CAD Late Mar-mid-Apr	13	15	0.9	0.7	80	Positive oil seasonality; potentially gov't fiscal year-end (31-Mar)
Late Oct/early Nov	43	44	-0.6	-0.5	73	Negative oil seasonality

* Median return, z-score and hit rate in this case is for EUR/USD rather than USD NEER (USD early May seasonality)

Source: J.P. Morgan

Figure 7: Swedish dividend outflows are a consistent driver of SEK weakness in the spring

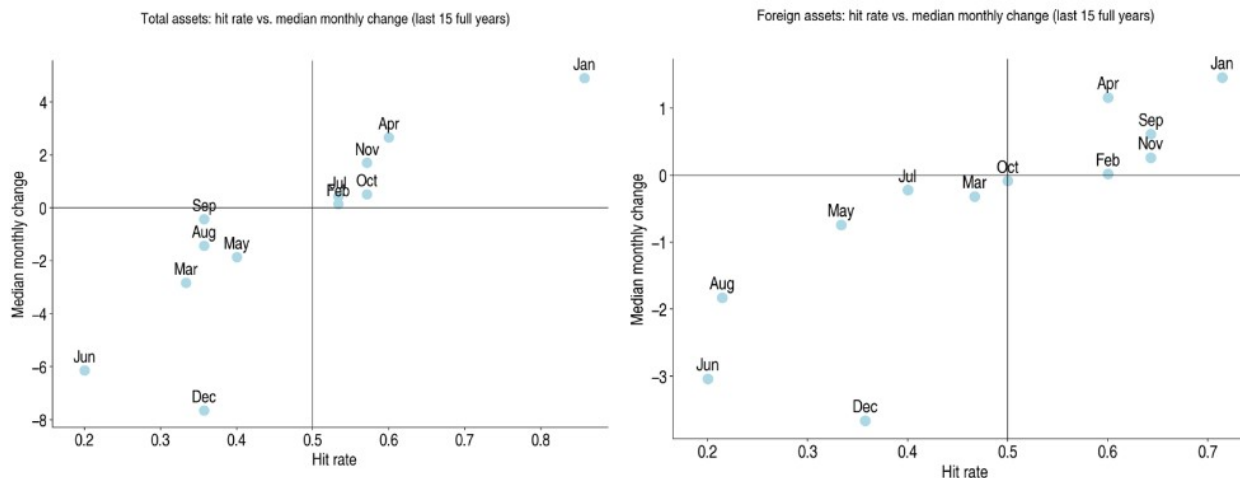
Dividend payouts as % of GDP (note around 50% are to foreign investors) by month (lhs) vs EUR/SEK indexed around peak dividend date (varies every year; usually in late March or April; rhs).



Dividend methodology following [Asia FX dividend seasonality](#), M. Loh
Source: J.P. Morgan

Figure 8: CHF strength in June and December and subsequent weakness in January can be reconciled with significant moves in

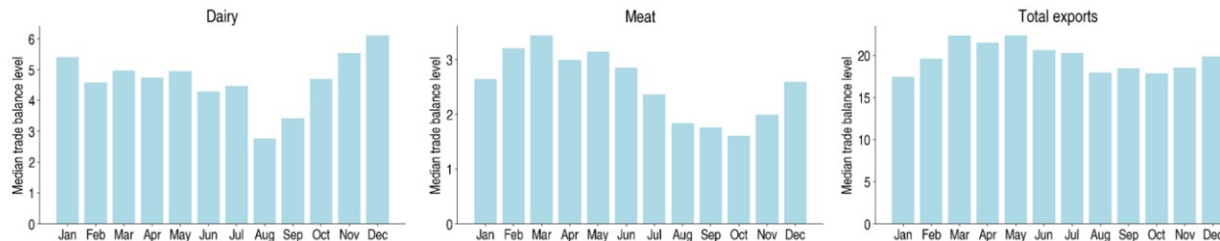
Median monthly change in total and foreign deposits in Swiss banks and hit rate (% of years where the change was in line with the median direction); 2011-2025.



Source: Haver, J.P. Morgan

Figure 9: NZD strength in December aligns with the seasonal peak in the dairy trade balance; Aug weakness with the low

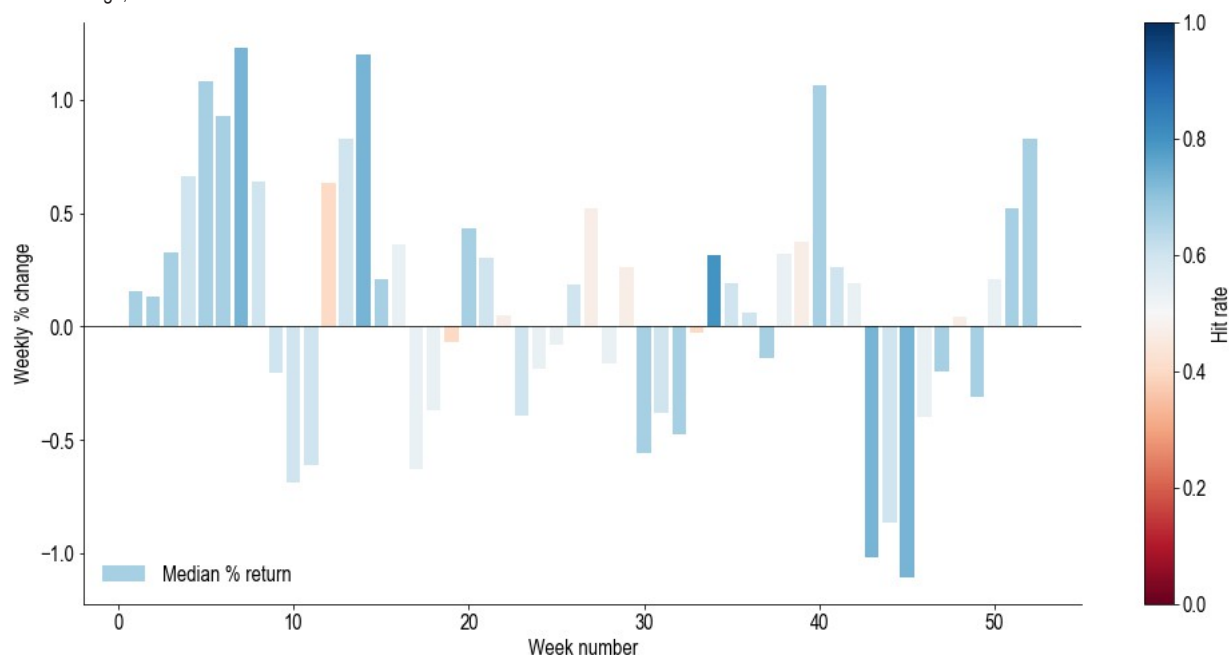
Median trade balance level by month as % of (monthly) GDP



Source: Haver, J.P. Morgan

Figure 10: Historical NOK & CAD weakness in Oct/Nov can be reconciled with negative oil seasonality over that period

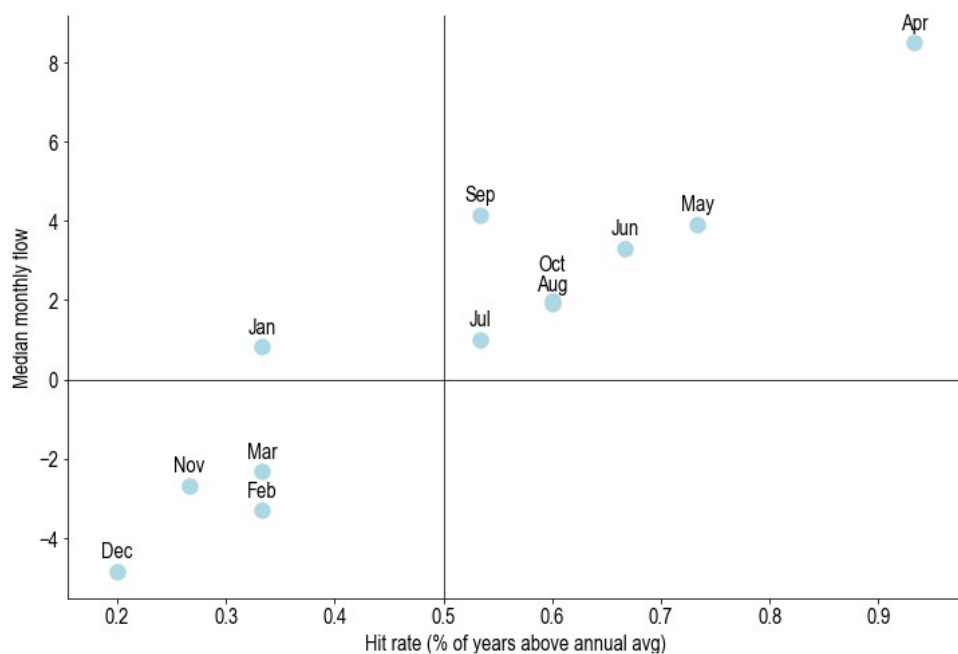
Tri-weekly smoothed median returns in Brent prices by week; colour reflects hit rate (% of years in which the return that week matches the direction of the median) and bar the median % change; 2011-2025



Source: Bloomberg L.P., J.P. Morgan

Figure 11: Lifer purchases of foreign debt in April used to weigh on JPY, but this is no longer the case since 2022

Lifer net acquisition of foreign long-term debt: hit rate vs median monthly flow (% of monthly GDP); 2011-2025



Source: Haver, J.P. Morgan

Part 2: A systematic approach to seasonality

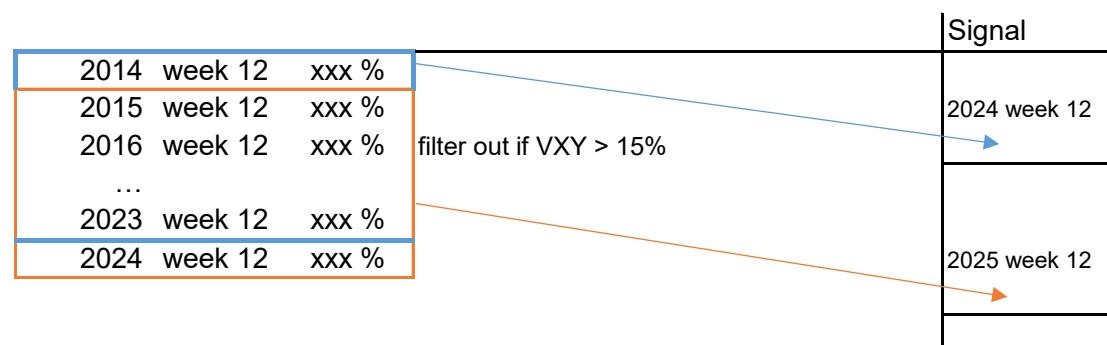
Our EM Asia FX strategists have previously published the [Asia FX seasonality guide](#) (M. Loh). In their analysis, weekly seasonality signals are generated by incorporating average tri-weekly returns for each currency and adjusted for each currency's idiosyncratic volatility. Our framework similarly applies a smoothing method across weeks and normalisation to enhance signal consistency. Together, this note and theirs provide a comprehensive catalogue of G10 and Asia currency seasonality patterns.

Methodology

The standardised seasonality index for a given currency is built as follows:

- 1. Data inputs:** We use either **CCY/USD** spot or **the Nominal Effective Exchange Rates (NEER)** as inputs to calculate seasonality starting from *January 1995*. We source CCY/USD spot data from DataQuery which contains data from our trading desk. NEERs are updated daily by our Economics team, which can also be found in DataQuery (*see [details](#) for computation methodology*). We consider our usual 20 most liquid and a global portfolio which contains 30 currencies in total.¹
- 2. Weekly returns:** FX returns are calculated per ISO 8601 week numbers (Sunday week-start, Monday–Friday business days per UK calendar), yielding ~52 weeks per year.
- 3. Standardisation:** Weekly returns are scaled by the trailing 1-year standard deviation to ensure cross-currency comparability.
- 4. Weekly seasonality metric construction:** For each week, we take the median of the standardised weekly return over the prior 10 years on a rolling basis, excluding the current year (e.g., week 12 of 2025 uses week 12 values from 2015–2024; the resulting dataset starts in 2004). Weeks where VXY Global exceeded 15% are excluded to avoid distortions from one-off market moves.

Figure 12: Illustration of weekly seasonality computation



Source: J.P. Morgan

- 5. Smoothing:** We evaluate four weighting schemes across adjacent weeks to capture potential spillover/forward-looking effects: 1) **Average weight on past/current/next week**; 2) **50% weight on current week; 25% weight on past & next week**; 3) **50%/50% weight on current/next week**; 4) **50%/50% weight on past/current week** (Figure 13). Applied to both CCY/USD and NEER inputs, this yields **8 signals** in total.

1. 30 currencies include: EUR, GBP, AUD, NZD, USD, CHF, CAD, NOK, SEK, JPY, BRL, CLP, CNY, HUF, INR, MXN, RUB, SGD, ZAR, COP, CZK, IDR, KRW, PHP, PLN, THB, TWD, ILS, TRY, MYR

Figure 13: Four different weighting schemes to smooth weekly seasonalities

Scheme	1	2	3	4
	Average of 3 weeks	50% current week, 25%/25% prev/next week	50%/50% current/next week	50%/50% past/current week
week t - 2	-	-	-	-
week t - 1	33%	25%	-	50%
week t	33%	50%	50%	50%
week t + 1	33%	25%	50%	-
week t + 2	-	-	-	-

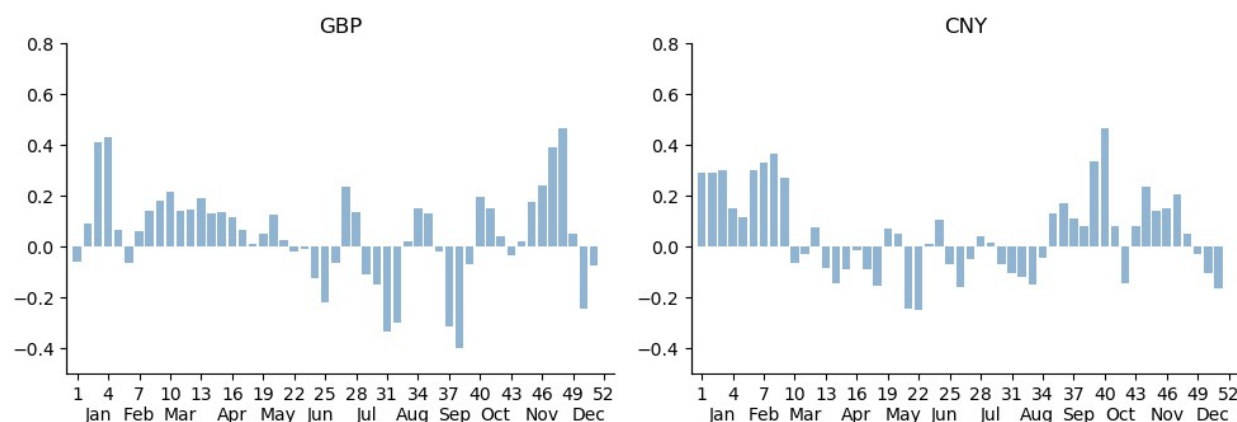
Source: J.P. Morgan

6. Cross-sectional backtest: is conducted on *G10*, *20 most liquid*, and *global* portfolios: buy/sell top/bottom (2/3/5) currencies vs. USD, equally weighted, rebalanced weekly on Monday.

For visualisation, we use the NEER-based signal with weighting scheme 2 (50% current, 25%/25% prev/next) as an example. Bar charts show the 2026 weekly seasonality signal reflecting 2016–2025 historical patterns; full currency details are in *Appendix A: Visualisation of seasonality signals*.

Figure 14: Examples of weekly seasonality signals for GBP & CNY per calendar week in 2026

Bar chart of weekly seasonality per calendar week – calculated based on NEER performance from 2016 - 2025 and applied the second weighting scheme: 50% current week, 25%/25% previous & next week



Source: J.P. Morgan

Backtesting results: Seasonality signal turned ineffective post-2020

Full performance statistics across all eight signals and three portfolios are in Appendix C, in Figure 26, Figure 27 and Figure 28, for the *G10*, *20 most liquid*, and *Global* portfolios, respectively. Key findings:

- **G10 portfolio: signals largely ineffective.** Best-performing signal is NEER + weighting **scheme (3)** (50%/50% current/next week), with a risk-adjusted return of only 0.15 ex-trading costs; most other variations produce negative risk-adjusted returns.
- **20 most liquid portfolio:** NEER + weighting **scheme (3)** delivers the best risk-adjusted returns of **0.4**.
- **Global portfolio:** NEER + **scheme (2)** delivers the best risk-adjusted return of **0.6**.

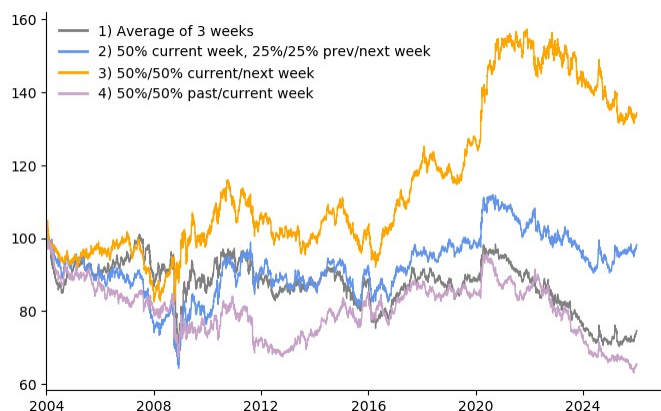
Consistent cross-portfolio findings:

1. **NEER-based signals consistently outperform CCY/USD signals**, with the gap most pronounced in the global portfolio (Figure 28), likely because CCY/USD returns embed a dollar seasonality component that contaminates idiosyncratic signals.

2. **Forward-looking weighting improves performance; backward-looking weighting is detrimental. Scheme (4) (50%-50% past-current week) is generally unprofitable even ex-costs** (risk-adjusted returns are negative; see Figure 27 and Figure 28). A plausible rationale is that if seasonality reflects a one-off event with limited macro impact (e.g., equity dividend outflows), price action from the prior week may already have reversed, making it a misleading signal.

Figure 15: G10 portfolio: seasonality signals are generally not effective

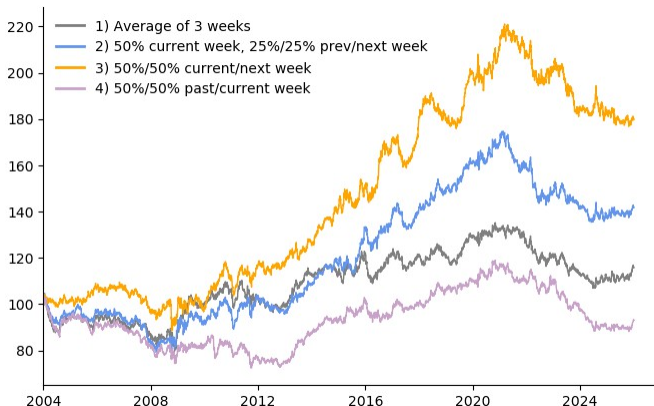
Cumulative spot returns of the 4 different seasonality signals based on NEER since 2004, top/bottom 2 on G10 portfolio, ex-trading costs



Source: J.P. Morgan

Figure 16: 20 most liquid portfolio: seasonality signals have delivered consistent performance since the GFC until around 2020-2021

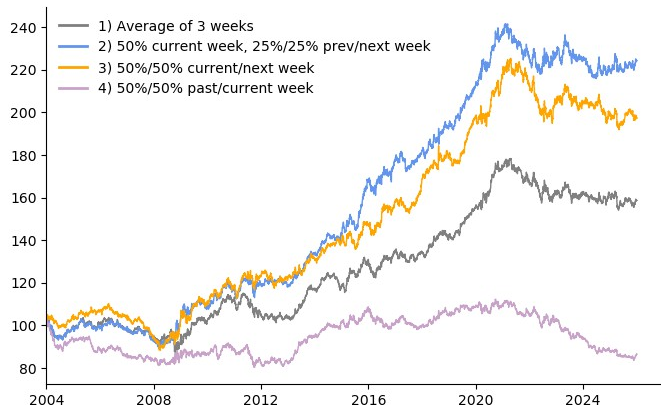
Cumulative spot returns of the 4 different seasonality signals based on NEER since 2004, top/bottom 3 on 20 most liquid currencies portfolio, ex-trading costs



Source: J.P. Morgan

Figure 17: Global portfolio: seasonality strategies yield better performance, but the strategy has similarly remained flat post 2020

Cumulative spot returns of the 4 different seasonality signals based on NEER since 2004, top/bottom 5 in the Global portfolio, ex-trading costs



Source: J.P. Morgan

However, **these strategies have not delivered consistent returns over time**. Seasonality baskets have broadly delivered positive returns from 2009 to around 2020-2021, delivering risk-adjusted returns of 0.4-0.7 across different variations and portfolios. Since then, these strategies have started to underperform, with most portfolios and signal variations yielding negative risk-adjusted returns post-2020 (see Figure 27 and Figure 28 for details). Transaction costs further erode performance, pushing several variations to near-zero or negative after costs (Figure 27). Despite these findings for cross-sectional systematic strategies, we see continued merit in monitoring FX seasonality as an input to macro views on specific currencies or as a feature within multi-signal strategies.

Isolating currency seasonality from S&P 500/interest rate dynamics

To strip out broad macro effects, we regress weekly NEER returns on S&P 500 returns and treat the residuals as the idiosyncratic component, then apply the same seasonality transformation to these residuals using our baseline weighting (50% current week, 25% prior week, 25% next week). For example, **NOK's equities-adjusted signal is more negative in weeks 47–48** (late November) than the baseline, pointing to currency-specific headwinds even under supportive equity conditions. However, on a cross-sectional basis, including these controls does not materially improve risk-adjusted performance (Figure 19, Figure 29); the seasonality basket still underperforms post-2020.

Figure 18: NOK's baseline seasonality signal vs S&P 500 adjusted as of 2025

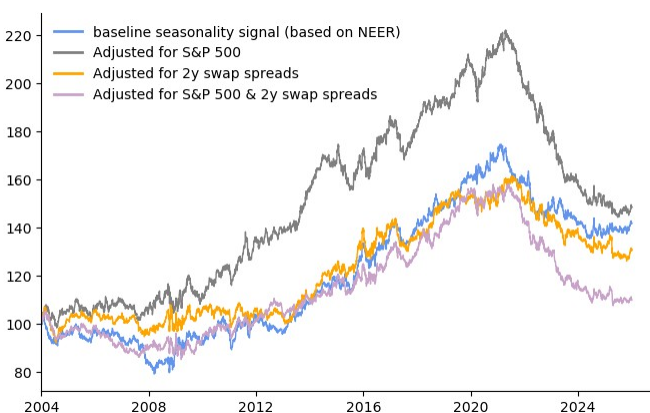
X axis: Baseline weekly seasonality signal based on NEER; Y axis: Residual seasonality signal (controlling for S&P 500)



Source: J.P. Morgan

Figure 19: Adjusting for seasonality in broader risk assets (proxied by S&P 500) and interest rate dynamics does not significantly improve strategy performance

Cumulative spot returns of the baseline seasonality signal (based on weekly NEER performance) vs. the residual seasonality signal adjusted for 1) S&P 500; 2) 2-year swap rate differential vs. USD; 3) both S&P 500 and 2-year swap rate differential vs. USD. Top/bottom 3 in 20 most liquid currencies portfolio, ex-trading costs



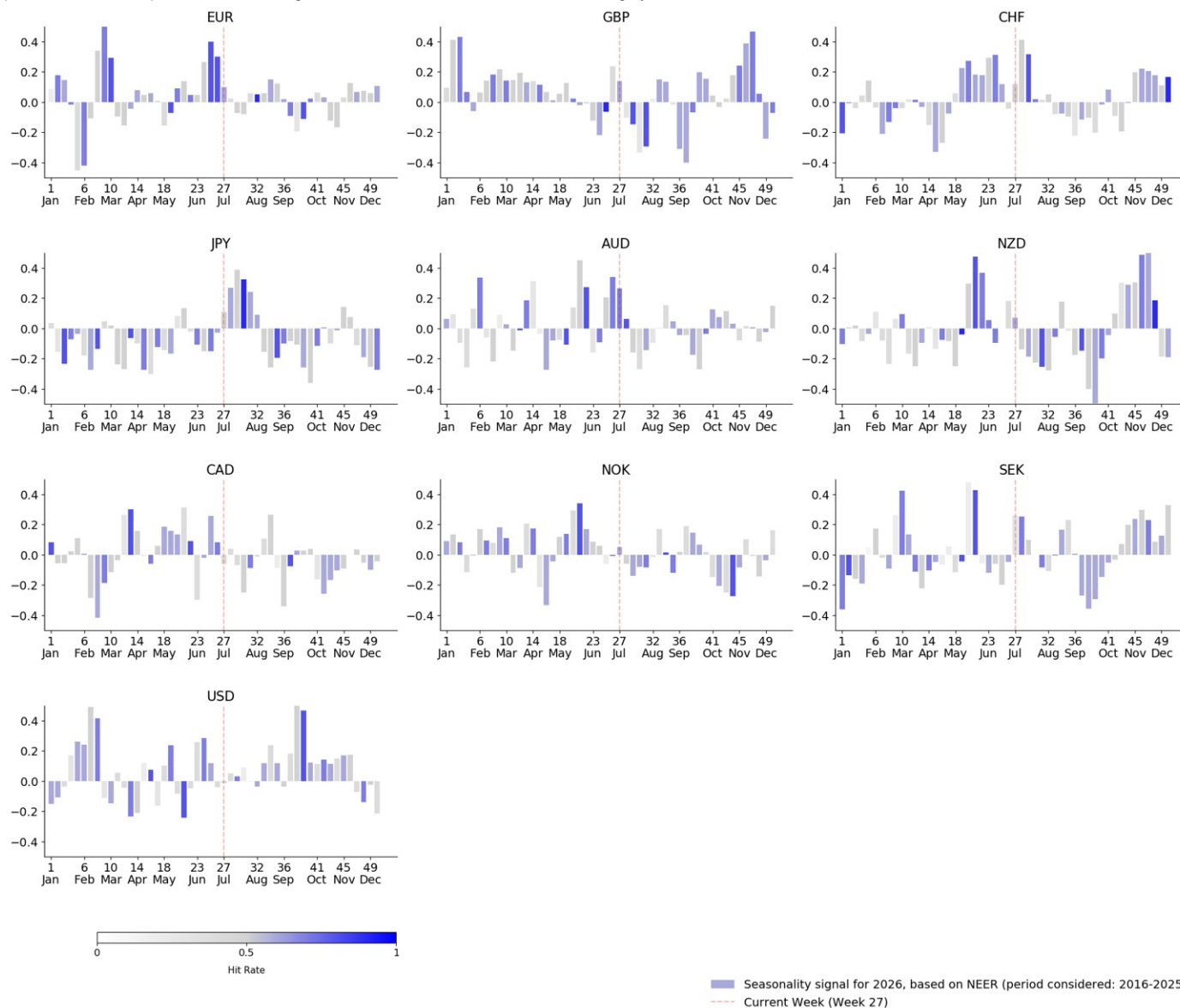
Source: J.P. Morgan

Appendix A: Visualisation of seasonality signals

Baseline seasonality signals based on NEER (50% weight on current week, 25% weight on past & next week)

Figure 20: G10 seasonality signals (NEER-based) per calendar week for 2026

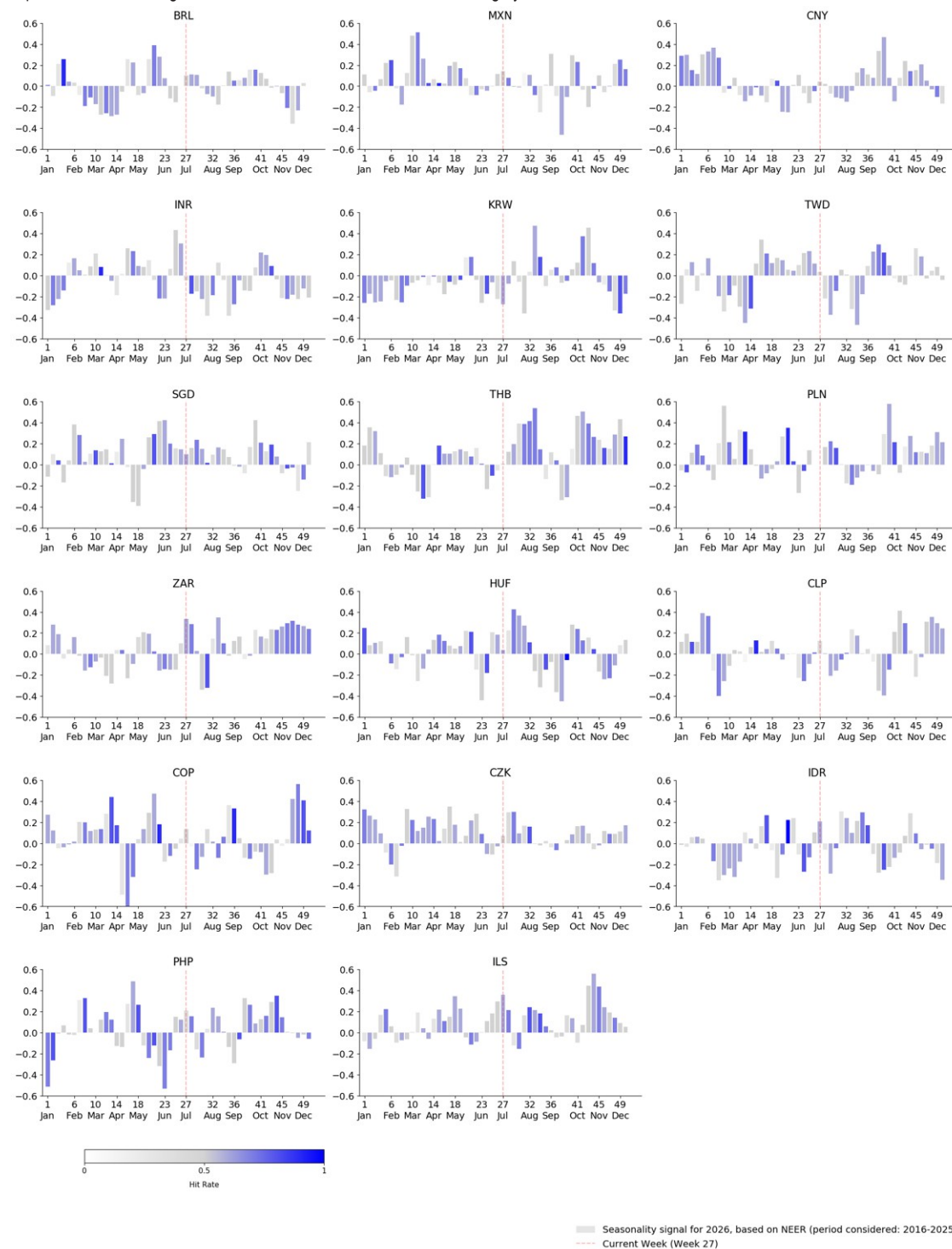
Bar chart of baseline weekly seasonality signal per calendar week -- calculated based on NEER performance from 2016 - 2025. Color scale indicates the hit rate of weekly NEER performance in the expected direction during 2016 - 2025: Blue indicates hit rate over 50%; grey indicates hit rate less than 50%



Source: J.P. Morgan

Figure 21: EM seasonality signals (NEER-based) per calendar week for 2026

Bar chart of weekly seasonality per calendar week - calculated based on NEER performance from 2016 - 2025. Color scale indicates hit rate of weekly NEER performance in the expected direction during 2016 - 2025: Blue indicates hit rate over 50%; grey indicates hit rate less than 50%

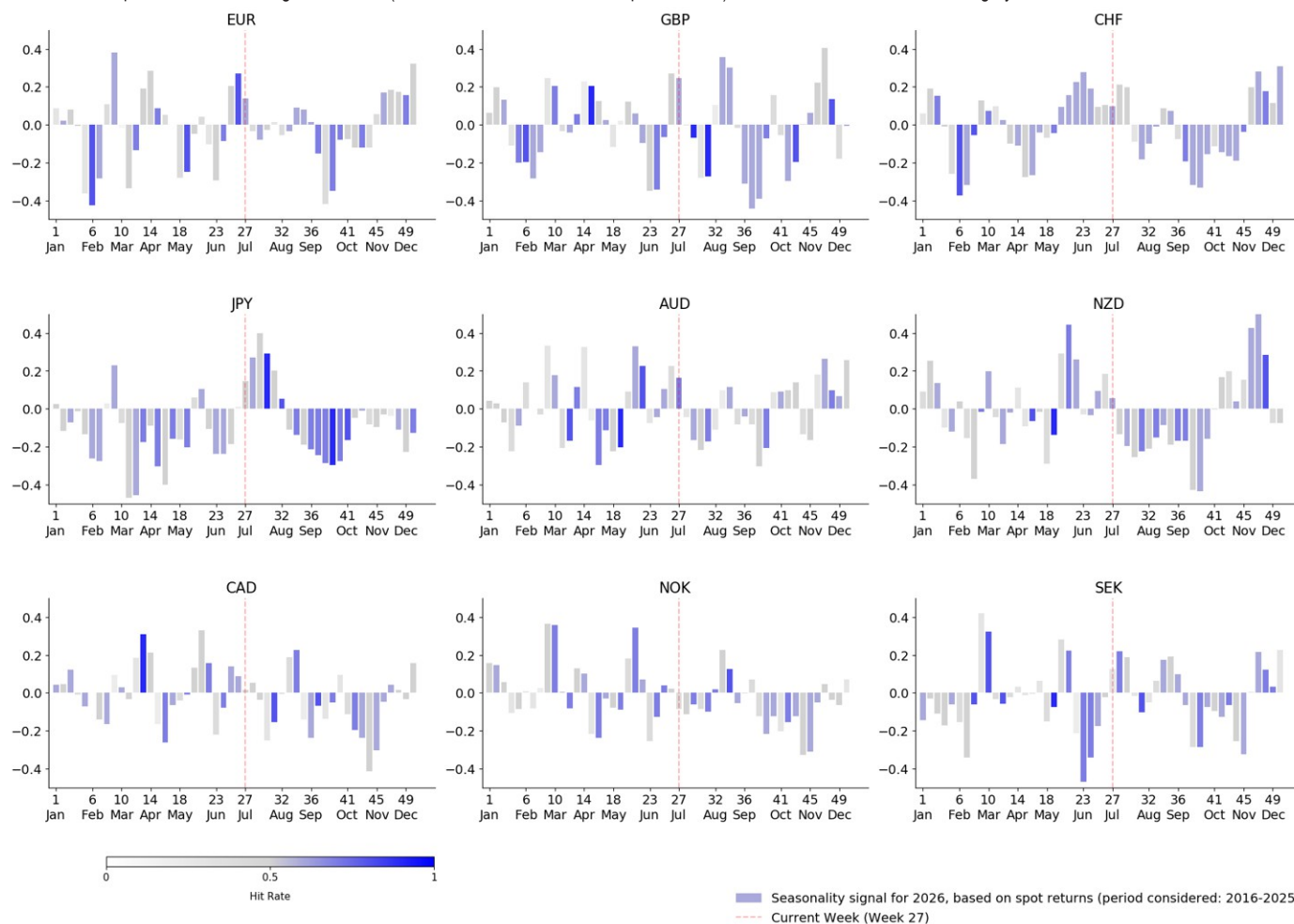


Source: J.P. Morgan

Baseline seasonality signals based on CCY/USD spot returns (50% weight on current week, 25% weight on past & next week)

Figure 22: G10 seasonality signals (CCY/USD spot based) per calendar week for 2026

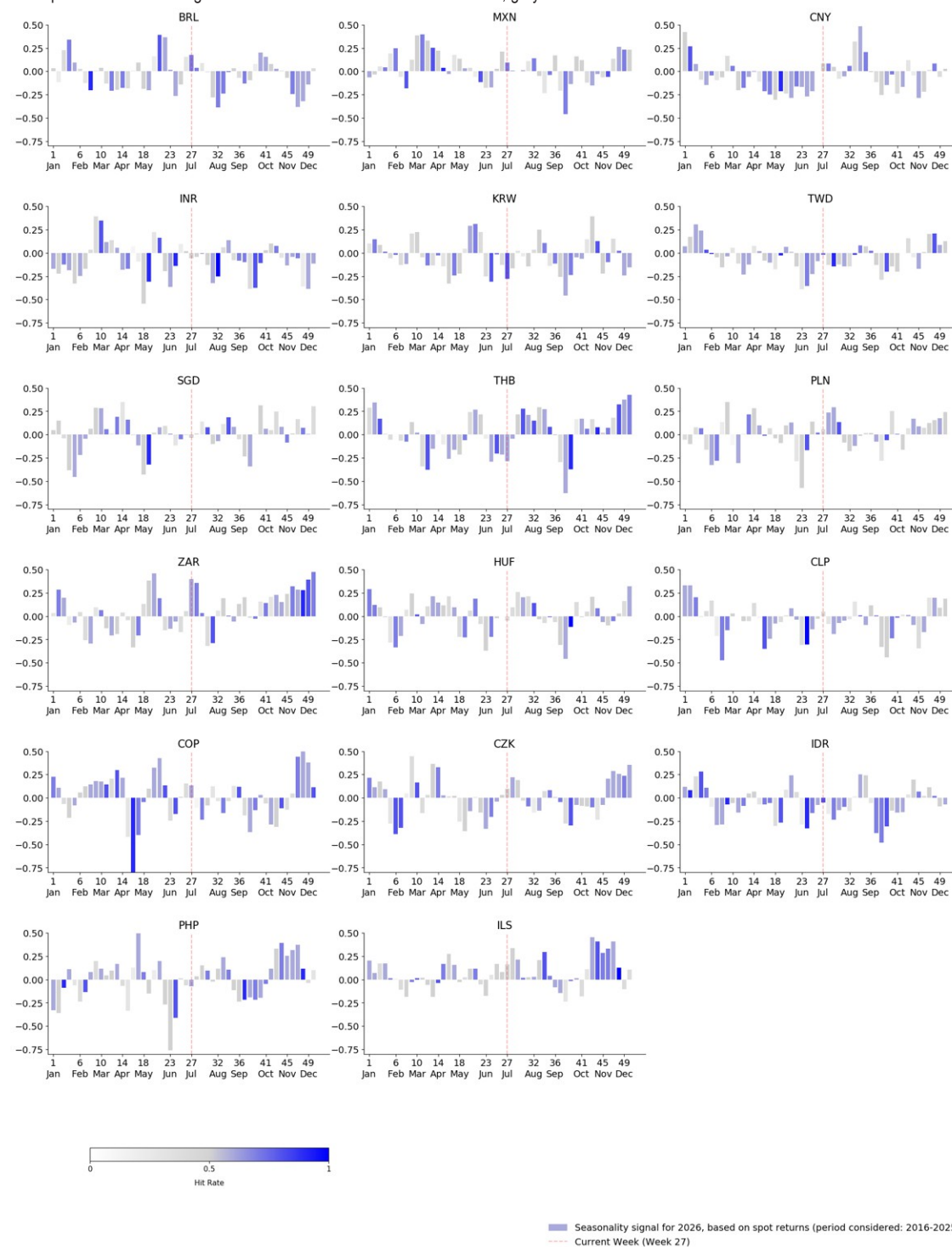
Bar chart of baseline weekly seasonality per calendar week - calculated based on CCY/USD performance from 2016 - 2025. Color scale indicates the hit rate of weekly CCY/USD returns in the expected direction during 2016 - 2025 (USD hit rate is based on NEER performance): Blue indicates hit rate over 50%; grey indicates hit rate less than 50%



Source: J.P. Morgan

Figure 23: EM seasonality signals (CCY/USD spot based) per calendar week for 2026

Bar chart of weekly seasonality per calendar week -- calculated based on CCY/USD performance from 2016 - 2025. Color scale indicates hit rate of weekly CCY/USD returns in the expected direction during 2016 - 2025: Blue indicates hit rate over 50%; grey indicates hit rate less than 50%



Source: J.P. Morgan

Appendix B: Hit rate of time-series seasonality strategy per currency pair

The following analysis compares hit rates of a time-series based strategy that buys/sells the currency pair if the seasonality signal is positive/negative, weekly rebalanced. A range of thresholds are also tested to account for the fact that seasonal patterns do not happen every week (in other words it would only trade seasonal patterns which are significant enough). Figure 24 and Figure 25 shows the results pre and post-Covid.

- Similar to the cross-sectional analysis, the time series results show a clear **regime split around 2020**. **Pre-2020**, seasonality signals were most effective for **GBP, NZD, NOK** in G10 (hit rates generally above 50% and rising with stricter thresholds), **CLP**, and in **Asia EM** – most notably **INR, PHP**, and **THB** (consistently >50% and improving as thresholds tighten). Several **G10 crosses** also stood out pre-2020 (e.g. EUR/JPY, AUD/CAD, CHF/JPY, NOK/SEK), with hit rates comfortably above 50% at higher thresholds.
- **Post 2020**, hit rates for **COP** and **CZK** have significantly improved compared to pre-2020, with hit rates consistently above 50% across different thresholds. Hit rates have also slightly improved for **SEK, HUF** and **ZAR** when **thresholds tighten**. For EM Asia, seasonality signals remain relevant for **INR** and **PHP**.

Figure 24: Seasonality signals works for some currencies before 2020...

Hit rate of time-series strategy from 2004-2020, which buy/sell CCY vs USD or CCY1 vs CCY2 when weekly seasonality signals exceed thresholds of 0, +/- 0.2, +/- 0.3, +/- 0.4, +/- 0.5. *CNH signals only starts in 2019. Seasonality signals are calculated using CCY/USD or CCY1/CCY2 as inputs, and smoothed by weighting **50% weight on current week; 25% weight on past & next week**), weekly rebalanced

CCY/USD pairs	threshold				
	0	0.2	0.3	0.4	0.5
G10:					
EUR/USD	47%	44%	44%	42%	44%
GBP/USD	52%	54%	56%	60%	55%
AUD/USD	50%	49%	47%	50%	46%
NZD/USD	50%	53%	55%	52%	42%
CHF/USD	49%	51%	51%	48%	43%
CAD/USD	51%	49%	47%	44%	41%
NOK/USD	52%	49%	53%	56%	54%
SEK/USD	48%	47%	48%	50%	53%
JPY/USD	54%	52%	51%	47%	45%
LatAm:					
CLP/USD	54%	57%	60%	60%	57%
COP/USD	49%	47%	52%	52%	35%
BRL/USD	51%	50%	46%	45%	43%
MXN/USD	50%	52%	54%	46%	42%
EMEA:					
PLN/USD	52%	53%	54%	50%	51%
CZK/USD	48%	45%	42%	51%	55%
HUF/USD	47%	48%	46%	46%	47%
ILS/USD	48%	45%	49%	50%	47%
ZAR/USD	50%	50%	52%	54%	55%
Asia:					
PHP/USD	51%	55%	56%	57%	59%
CNY/USD	54%	44%	50%		
INR/USD	53%	57%	59%	57%	56%
KRW/USD	50%	53%	54%	55%	57%
TWD/USD	52%	45%	47%	48%	45%
SGD/USD	49%	49%	52%	54%	53%
THB/USD	50%	49%	50%	56%	68%
frequently-tracked crosses					
EUR/GBP	52%	52%	51%	53%	54%
EUR/NOK	52%	52%	51%	53%	54%
EUR/SEK	48%	49%	48%	44%	36%
AUD/NZD	53%	50%	52%	53%	51%
NOK/SEK	51%	55%	54%	53%	52%
standout crosses					
EUR/JPY	53%	54%	56%	55%	70%
GBP/JPY	53%	55%	57%	56%	61%
AUD/CAD	51%	53%	53%	63%	66%
AUD/NOK	51%	53%	53%	55%	54%
CHF/JPY	53%	54%	54%	59%	63%

Source: J.P. Morgan

Figure 25: ...while these signals turned less relevant post 2020

Hit rate of time-series strategy from 2020-2025, which buy/sell CCY vs USD or CCY1 vs CCY2 when weekly seasonality signals exceed thresholds of 0, +/- 0.2, +/- 0.3, +/- 0.4, +/- 0.5. *CNH signals only starts in 2019. Seasonality signals are calculated using CCY/USD or CCY1/CCY2 as inputs, and smoothed by weighting **50% weight on current week; 25% weight on past & next week**), weekly rebalanced

CCY/USD pairs	threshold				
	0	0.2	0.3	0.4	0.5
G10:					
EUR/USD	50%	52%	54%	45%	29%
GBP/USD	50%	40%	48%	33%	43%
AUD/USD	46%	43%	39%	38%	100%
NZD/USD	46%	41%	35%	33%	25%
CHF/USD	49%	53%	52%	33%	25%
CAD/USD	50%	48%	44%	50%	100%
NOK/USD	46%	41%	38%	33%	14%
SEK/USD	53%	53%	54%	55%	33%
JPY/USD	50%	49%	46%	47%	38%
LatAm:					
CLP/USD	47%	46%	49%	46%	50%
COP/USD	51%	53%	56%	64%	67%
BRL/USD	47%	41%	39%	41%	63%
MXN/USD	51%	45%	46%	50%	55%
EMEA:					
PLN/USD	52%	51%	45%	48%	40%
CZK/USD	51%	54%	51%	54%	67%
HUF/USD	49%	51%	47%	88%	100%
ILS/USD	56%	48%	53%	50%	50%
ZAR/USD	52%	45%	53%	63%	100%
Asia:					
PHP/USD	52%	52%	51%	60%	63%
CNY/USD	52%	46%	42%	31%	33%
INR/USD	53%	58%	61%	50%	50%
KRW/USD	48%	45%	56%	32%	33%
TWD/USD	51%	43%	52%	38%	
SGD/USD	46%	48%	51%	60%	50%
THB/USD	46%	42%	42%	48%	53%
frequently-tracked crosses					
EUR/GBP	50%	49%	50%	54%	100%
EUR/NOK	52%	44%	44%	47%	
EUR/SEK	47%	48%	43%	33%	17%
AUD/NZD	51%	54%	54%	51%	50%
NOK/SEK	51%	50%	48%	59%	50%
standout crosses					
EUR/JPY	46%	36%	26%	27%	33%
GBP/JPY	51%	39%	50%	45%	14%
AUD/CAD	47%	38%	40%	44%	45%
AUD/NOK	48%	47%	46%	64%	80%
CHF/JPY	50%	50%	50%	55%	38%

Source: J.P. Morgan

Appendix C: Summary statistics of cross-sectional seasonality strategy

Figure 26: G10 portfolio: seasonality signals have generally not worked for the G10 portfolio

Summary performance statistics of different seasonality signals for the G10 portfolio. (top/bottom 2).

Period: 2004 - 2025	1		2		3		4	
Scheme	Average of 3 weeks		50% current week, 25%/25% prev/next week		50%/50% current/next week		50%/50% past/current week	
Signal	CCY/USD	NEER	CCY/USD	NEER	CCY/USD	NEER	CCY/USD	NEER
Seasonality measure								
Annualized returns	-2.8%	-2.4%	-1.4%	-1.2%	-0.7%	0.3%	-3.3%	-3.0%
Annualized vol	8.8%	8.5%	8.6%	8.5%	8.8%	8.3%	8.8%	8.4%
Risk-adj returns (with TC)	-0.32	-0.28	-0.17	-0.14	-0.08	0.03	-0.37	-0.36
Risk-adj returns (no TC)	-0.22	-0.16	-0.08	-0.03	0.03	0.15	-0.26	-0.23
Risk-adj pre 2020	-0.16	-0.08	-0.06	-0.01	-0.02	0.17	-0.08	-0.11
Risk-adj post 2020	-0.40	-0.45	-0.16	-0.11	0.17	0.09	-0.90	-0.61
Annualized TC	0.5%	0.5%	0.5%	0.5%	0.6%	0.6%	0.6%	0.6%
Max DD (6 months)	-28%	-28%	-21%	-22%	-17%	-15%	-26%	-21%
Correlation to G10 Risk-adj carry (monthly)	0.30	0.36	0.30	0.34	0.27	0.28	0.30	0.31
Correlation to USD TWI (monthly)	-0.14	-0.13	-0.15	-0.12	-0.13	-0.10	-0.16	-0.13

Universe: G10; Portfolio weighting: top/bottom 2; Rebalancing: weekly

G10 currencies: EUR, GBP, AUD, NZD, CHF, CAD, NOK, SEK, JPY, USD

Source: J.P. Morgan

Figure 27: 20 most liquid currency portfolio: weighting more on seasonality for the next week improves strategy returns, while inclusion of previous week's seasonality has worsened the strategy's performance

Summary performance statistics of different seasonality signals for the 20 most liquid currencies portfolio (top/bottom 3). Baseline seasonality strategy highlighted in red.

Period: 2004 - 2025	1		2		3		4	
Scheme	Average of 3 weeks		50% current week, 25%/25% prev/next week		50%/50% current/next week		50%/50% past/current week	
Weighting rules	CCY/USD	NEER	CCY/USD	NEER	CCY/USD	NEER	CCY/USD	NEER
Seasonality measure								
Annualized returns	-1.0%	-1.2%	-0.5%	-0.3%	0.5%	0.4%	-2.6%	-2.4%
Annualized vol	7.7%	7.5%	7.7%	7.6%	7.8%	7.6%	7.6%	7.5%
Risk-adj returns (with TC)	-0.13	-0.15	-0.07	-0.03	0.07	0.06	-0.34	-0.32
Risk-adj returns (no TC)	0.13	0.09	0.19	0.21	0.35	0.35	-0.05	-0.05
Risk-adj returns pre 2020 (no TC)	0.32	0.20	0.37	0.37	0.44	0.54	0.08	0.08
Risk-adj returns post 2020 (no TC)	-0.45	-0.25	-0.35	-0.27	0.06	-0.23	-0.45	-0.39
Annualized TC	1.1%	1.1%	1.2%	1.1%	1.3%	1.3%	1.3%	1.4%
Max DD (6 months)	-16%	-15%	-18%	-15%	-16%	-14%	-14%	-17%
Correlation to Global Risk-adj carry (monthly)	0.07	0.11	0.08	0.11	0.07	0.11	0.06	0.11
Correlation to USD TWI (monthly)	-0.10	-0.07	-0.08	-0.08	-0.08	-0.06	-0.10	-0.09

Universe: 20 Liquid ccy; Portfolio weighting: top/bottom 3; Rebalancing: weekly

20 most liquid currencies: EUR, GBP, AUD, NZD, CHF, CAD, NOK, SEK, JPY, USD, BRL, MXN, PLN, ZAR, CNY, INR, KRW, TWD, SGD, THB

Source: J.P. Morgan

Figure 28: Global portfolio: using NEERs as seasonality inputs improves strategy returns significantly compared to using CCY/USD returns

Summary performance statistics of different seasonality signals for the Global portfolio (top/bottom 5). Baseline seasonality strategy highlighted in red.

Period: 2004 - 2025	1		2		3		4	
Scheme	Average of 3 weeks		50% current week, 25%/25% prev/next week		50%/50% current/next week		50%/50% past/current week	
Signal	CCY/USD	NEER	CCY/USD	NEER	CCY/USD	NEER	CCY/USD	NEER
Seasonality measure								
Annualized returns	-0.4%	-0.4%	-0.8%	1.0%	-0.6%	0.0%	-3.3%	-3.8%
Annualized vol	6.1%	6.1%	6.2%	6.1%	6.2%	6.2%	6.0%	6.1%
Risk-adj returns (with TC)	-0.06	-0.07	-0.13	0.16	-0.10	0.00	-0.54	-0.62
Risk-adj returns (no TC)	0.32	0.34	0.23	0.59	0.35	0.50	-0.10	-0.11
Risk-adj pre 2020	0.43	0.44	0.43	0.74	0.37	0.66	0.08	0.08
Risk-adj post 2020	0.01	0.07	-0.30	0.13	0.27	-0.01	-0.62	-0.65
Annualized TC	1.4%	1.5%	1.4%	1.6%	1.7%	1.9%	1.7%	1.9%
Max DD (6 months)	-12%	-11%	-14%	-12%	-15%	-13%	-14%	-14%
Correlation to Global Risk-adj carry (monthly)	0.09	0.12	0.09	0.12	0.09	0.12	0.06	0.12
Correlation to USD TWI (monthly)	-0.12	-0.11	-0.10	-0.08	-0.08	-0.07	-0.10	-0.09

Universe: Global; Portfolio weighting: top/bottom 5; Rebalancing: weekly

Global currencies: EUR, GBP, AUD, NZD, CHF, CAD, NOK, SEK, JPY, USD, BRL, MXN, PLN, ZAR, CNY, INR, KRW, TWD, SGD, THB, CZK, HUF, IDR, MYR, PHP, RUB, TRY, ILS, CLP, COP

Source: J.P. Morgan

Figure 29: Adjusting for seasonality in broader risk assets or relative rate spreads does not necessarily improve strategy performance

Summary performance statistics of our Baseline seasonality signal (based on weekly NEER performance) vs. Residual seasonality signals adjusted for 1) S&P 500; 2) 2-year swap rate differentials vs. USD; 3) both S&P 500 and 2-year swap rate differentials vs. USD, for the 20 most liquid currencies portfolio.

Signal	NEER based & Scheme 2) weighting	Adj for S&P 500	Adj for 2y swap spreads	Adj for S&P 500 + 2y swap spreads
Seasonality measure	NEER	NEER	NEER	NEER
Annualized returns	-0.3%	-0.2%	0.1%	-0.6%
Annualized vol	7.6%	7.7%	7.3%	7.1%
Risk-adj returns	-0.03	-0.03	0.02	-0.08
Risk-adj returns (no TC)	0.21	0.23	0.17	0.08
Annualized TC	1.1%	1.8%	0.9%	0.9%
Max DD (6 months)	-15%	-12%	-14%	-14%
Correlation to Global Risk-adj carry (monthly)	0.11	0.08	0.09	0.07
Correlation to USD TWI (monthly)	-0.08	-0.10	-0.07	-0.08

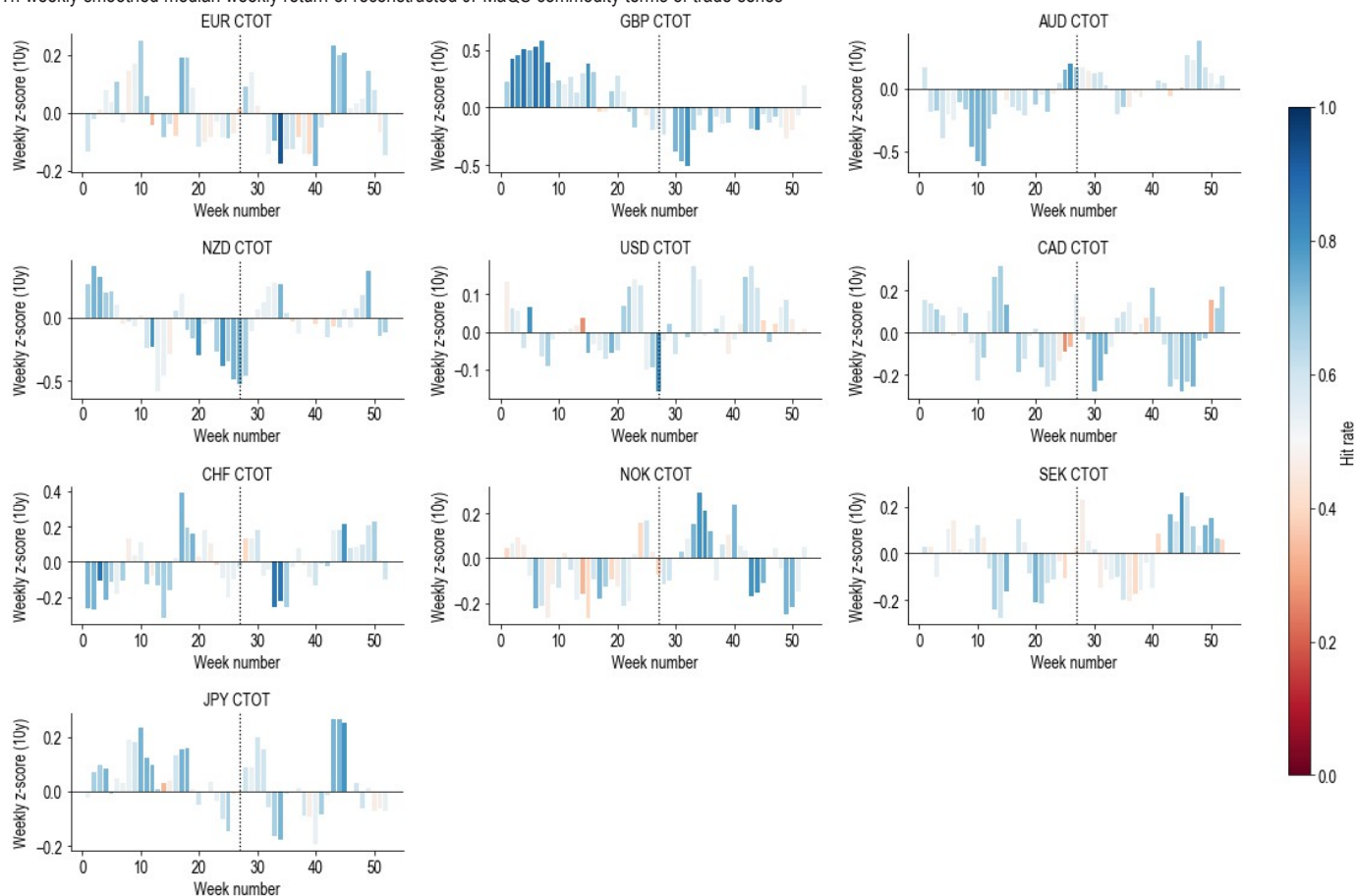
Universe: 20 Liquid ccy; Portfolio weighting: top/bottom 3; Rebalancing: weekly

20 most liquid currencies: EUR, GBP, AUD, NZD, CHF, CAD, NOK, SEK, JPY, USD, BRL, MXN, PLN, ZAR, CNY, INR, KRW, TWD, SGD, THB
Source: J.P. Morgan

Appendix D

Figure 30: Commodity terms of trade seasonality

Tri-weekly smoothed median weekly return of reconstructed JPMaQS commodity terms of trade series



Source: JPMaQS, J.P. Morgan

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Completed 03 Jul 2026 04:09 PM BST

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